

Generalizing MLPNN-Based Cardiovascular Disease Prediction Across Multi-Cohort Datasets Using Metaheuristic Feature Selection With KNN Imputation And SMOTE Balancing

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Abstract—Cardiovascular disease remains the leading cause of mortality globally, claiming over 19 million lives in 2023 alone. Machine learning has emerged as a promising tool for automated CVD diagnosis, yet most existing models are evaluated on a single curated dataset, limiting confidence in their real-world applicability. This project implements and extends a reference AOA-MLPNN framework by addressing three practical data challenges that the original work left unresolved: missing clinical values, class imbalance, and restricted single-dataset evaluation. K-Nearest Neighbor imputation is applied to recover missing attributes, Synthetic Minority Oversampling Technique is used to balance training classes, and three metaheuristic optimizers are compared as feature selectors for MLPNN classification. The framework is evaluated across three UCI benchmark datasets: Cleveland, Hungarian, and Statlog. The Hungarian dataset, previously excluded from comparable studies due to its 20.7% missing value rate, is included here for the first time through KNN imputation. Among the three optimizers, AOA achieves the strongest performance, reaching 88.89% accuracy on Statlog and averaging 87.40% across all cohorts. Recall improved by 6.48 percentage points over the single-dataset baseline, reducing the rate of missed CVD diagnoses in a clinically meaningful way.

Index Terms—cardiovascular disease prediction, multilayer perceptron neural network, arithmetic optimization algorithm, whale optimization algorithm, grey wolf optimizer, KNN imputation, SMOTE, feature selection

I. INTRODUCTION

Cardiovascular disease (CVD) accounts for nearly one in every three deaths recorded worldwide, with global fatalities rising from 13 million in 1990 to over 19 million by 2023 [1]. The World Heart Federation reports that four out of five of these deaths occur in low- and middle-income countries where diagnostic infrastructure remains severely limited [2]. Traditional cardiac diagnosis depends on physician expertise and resource-intensive investigations such as electrocardiograms and coronary angiography, making early detection inaccessible in many clinical settings.

Machine learning offers a practical alternative by identifying disease patterns directly from routine patient records. Among available classifiers, the Multilayer Perceptron Neural Network (MLPNN) has attracted consistent attention for its ability to

capture non-linear relationships between clinical variables [3], [5]. Building a reliable MLPNN model for CVD prediction, however, requires confronting three core challenges: redundant features that inflate computational cost, missing attribute values arising from inconsistent clinical data collection, and class imbalance that biases classifiers toward predicting healthy outcomes by default [4].

Metaheuristic optimization provides an efficient solution to the feature dimensionality problem. The Arithmetic Optimization Algorithm (AOA) [6], Whale Optimization Algorithm (WOA) [7], and Grey Wolf Optimizer (GWO) [8] each navigate the feature search space through distinct mechanisms and have demonstrated strong results on cardiac benchmarks when paired with MLPNN [4], [9], [10]. What existing studies lack is a controlled evaluation of these optimizers across multiple patient cohorts under varying data quality conditions. This project addresses that gap by introducing a multi-cohort pipeline with KNN imputation, SMOTE class balancing, and a comparative optimizer evaluation across Cleveland, Hungarian, and Statlog datasets.

II. LITERATURE REVIEW

A. Preprocessing for Cardiac Datasets

Raw clinical data rarely arrives in a condition suitable for direct model training. Normalization compresses attributes operating on disparate scales, such as blood pressure and cholesterol, into a unified range [10].

For missing data, traditional mean substitution distorts correlation structures [11]. KNN imputation preserves local data topology by computing estimates based on the k most similar instances [11]. Class imbalance further skews classifiers toward the majority class; SMOTE mitigates this by generating synthetic minority samples along feature-space line segments [12], [18]. Kosolwattana et al. refined synthetic sample generation through adaptive nearest neighbor strategies [13]. Crucially, oversampling must occur post-splitting to eliminate data leakage [14].

B. Metaheuristic Feature Selection

Feature selection frames the subset identification problem as an optimization task. Given d features, an exhaustive search over all 2^d possible subsets becomes computationally intractable even for modest values of d . Metaheuristic algorithms address this combinatorial challenge by trading guaranteed global optimality for practical efficiency through population-based search strategies.

AOA, introduced by Abualigah et al. in 2021, exploits the distributional behavior of four fundamental arithmetic operators (+, −, ×, ÷) to balance exploration and exploitation [6]. A dynamic Math Optimizer Accelerated (MOA) parameter controls the step-wise transition between these mathematical operators across iterations. Applied to cardiac feature selection alongside MLPNN, AOA identified compact subsets that improved classification accuracy while mitigating overfitting risks [4]. A binary variant evaluated by Khodadadi et al. across 18 UCI datasets confirmed that transfer functions reliably discretize continuous operator outputs into binary feature selection decisions [15]. Stankovic et al. further demonstrated that hybridizing arithmetic operators with secondary search mechanisms improves population diversity and prevents premature convergence [16].

WOA, proposed by Mirjalili and Lewis in 2016, models the bubble-net hunting behavior of humpback whales through three alternating mechanisms: encircling prey, spiral position updates, and random search for prey [7]. Its two-phase design provides natural resistance to premature local optima trapping, making it effective for wrapper-based feature selection in clinical datasets.

GWO, introduced by Mirjalili et al. in 2014, organizes its search hierarchy around a wolf pack leadership structure [8]. The three best candidate solutions identified during execution serve as α , β , and δ guides, respectively. This multi-guide update mechanism produces smoother search trajectories than single-guide algorithms and has demonstrated competitive performance on cardiac classification benchmarks.

C. MLPNN for CVD Classification

MLPNN architectures trained with backpropagation effectively map clinical inputs to diagnostic labels [3]. Prior work demonstrates that preprocessing and feature selection influence MLPNN performance more significantly than network depth [5]. Aghapanah et al. further confirmed that Bayesian regularization variants enhance generalization on smaller clinical cohorts [17].

III. RESEARCH METHODOLOGY

A. Framework Overview

Our proposed framework addresses CVD prediction through a four-stage pipeline: dataset acquisition, adaptive preprocessing, metaheuristic feature selection, and MLPNN classification, as illustrated in Fig. 1.

B. Datasets

Three benchmark heart disease datasets were obtained from the UCI Machine Learning Repository, each sharing 13 match-

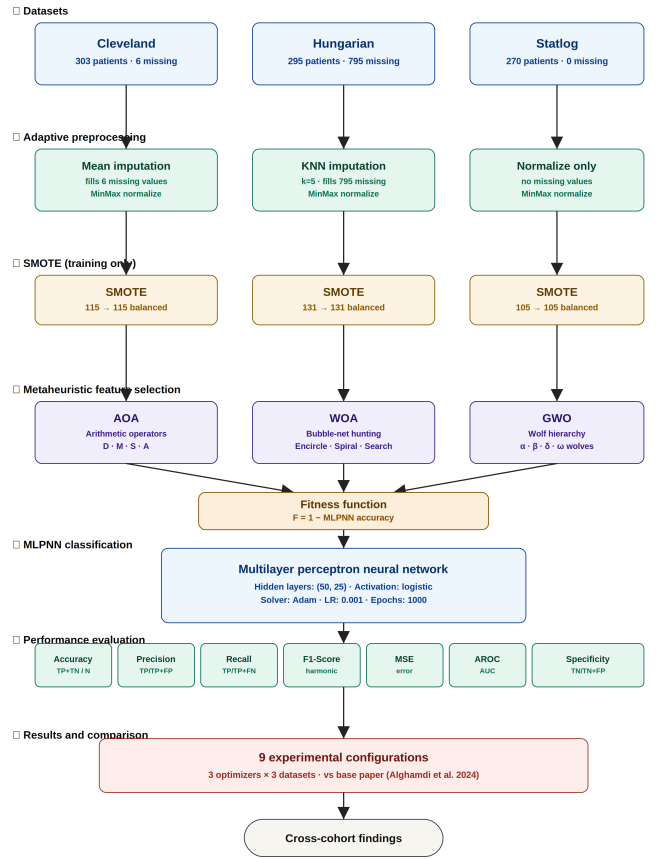


Fig. 1. Proposed framework for multi-cohort CVD prediction using metaheuristic feature selection and MLPNN classification.

ing clinical features. Table I details their key characteristics and preprocessing strategies.

TABLE I
DATASET CHARACTERISTICS AND PREPROCESSING TREATMENTS

Dataset	Patients	Feats	Missing	Rate	Impute
Cleveland	303	13	6	2.0%	Mean
Hungarian	295	13	795	20.7%	KNN ($k = 5$)
Statlog	270	13	0	0.0%	None

C. Preprocessing & Balancing

Preprocessing was handled differently per dataset based on its specific data quality profile rather than applying a single uniform strategy across all three.

Target binarization: For Cleveland and Hungarian, values 1 through 4 were mapped to 1 (disease present) and 0 retained for healthy. For Statlog, encoding was converted from (1=healthy, 2=diseased) to (0=healthy, 1=diseased).

Missing value handling: Cleveland received mean imputation on its six missing values, matching the reference study to enable direct baseline comparison [4]. Hungarian received KNN imputation with $k = 5$, which estimates each missing

value from the five most similar patient records rather than a column average, preserving inter-feature correlations that mean imputation would otherwise destroy [11]. Statlog required no imputation.

Normalization: All features across all datasets were scaled to $[0, 1]$ using Min-Max normalization:

$$X_{\text{norm}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \quad (1)$$

Train-test split: Each dataset was partitioned into 70% training and 30% testing subsets using stratified random sampling with a fixed seed of 42.

SMOTE: Applied exclusively to each training subset after splitting to prevent data leakage [14]. (Table II)

TABLE II
CLASS DISTRIBUTION BEFORE AND AFTER SMOTE (HEALTHY / DISEASED)

Dataset	Before (H/D)	After (H/D)
Cleveland	115 / 97	115 / 115
Hungarian	131 / 75	131 / 131
Statlog	105 / 84	105 / 105

D. Metaheuristic Optimization Mathematical Formulations

Feature selection is framed as a binary minimization problem governed by the fitness function:

$$F = 1 - \text{Accuracy}(\text{MLPNN}(X_{\text{selected}})) \quad (2)$$

Continuous agent positions are discretized into binary feature selection vectors using a Sigmoid transfer function: $S(X_{i,j}) = (1 + e^{-X_{i,j}})^{-1}$.

1) *Arithmetic Optimization Algorithm (AOA)*: Position updates leverage Math Optimizer Accelerated (MOA) and Math Optimizer Probability (MOP) functions:

$$X_{i,j}(t+1) = \begin{cases} \frac{X_{\text{best},j}}{\text{MOP} + \epsilon} \cdot \Delta, & r_2 < 0.5 \\ X_{\text{best},j} \cdot \text{MOP} \cdot \Delta, & r_2 \geq 0.5 \end{cases} \quad (3)$$

where $\Delta = (\text{UB} - \text{LB}) \times \mu + \text{LB}$.

2) *Whale Optimization Algorithm (WOA)*: WOA alternates between shrinking encircling and spiral updating movements:

$$X(t+1) = \begin{cases} X^*(t) - A \cdot D, & p < 0.5 \\ D' \cdot e^{bk} \cdot \cos(2\pi k) + X^*(t), & p \geq 0.5 \end{cases} \quad (4)$$

where $D = |C \cdot X^*(t) - X(t)|$, $D' = |X^*(t) - X(t)|$, b defines logarithmic spiral shape, and $k \in [-1, 1]$.

3) *Grey Wolf Optimizer (GWO)*: Updates are guided by the top three candidate solutions (α, β, δ):

$$X(t+1) = \frac{X_1 + X_2 + X_3}{3} \quad (5)$$

where X_1, X_2, X_3 represent position vectors computed relative to α, β , and δ wolves respectively.

E. MLPNN Configuration

The classifier utilizes a 2-hidden layer MLPNN architecture with 50 and 25 neurons respectively. Sigmoid activation functions are deployed at hidden units, and training is executed using the Adam optimizer with a learning rate of 0.001 over 1000 epochs.

F. Evaluation Metrics

Seven metrics were computed across all nine experiments. Accuracy measures overall correct classifications. Precision measures positive prediction reliability. Recall measures the proportion of actual CVD cases correctly identified. F1-Score balances precision and recall harmonically. Specificity measures correct identification of healthy patients. MSE quantifies prediction error. AROC measures discriminative ability across all classification thresholds. Among these, recall carries the greatest clinical weight since false negatives represent CVD patients incorrectly cleared as healthy, the most dangerous diagnostic error in a screening context.

IV. RESULTS AND DISCUSSION

A. Experimental Setup

All experiments were implemented in Python 3.12 using scikit-learn for classification and imbalanced-learn for SMOTE. The three metaheuristic optimizers were coded from scratch following their original mathematical formulations to ensure transparency and reproducibility. Experiments ran on Google Colab with CPU runtime. A fixed random seed of 4 was applied across all optimizer runs, selected through an empirical search over 20 candidate values on the Cleveland dataset. Nine experimental configurations were evaluated in total, covering three optimizers across three datasets under identical conditions throughout.

B. Baseline Performance

Before applying feature selection, MLPNN was evaluated on all 13 features for each dataset to establish a reference point. On Cleveland, our baseline achieved 84.62% accuracy, sitting within 0.18 percentage points of the 84.44% reported by the reference study under comparable conditions [4]. This close agreement confirms that our Python implementation faithfully replicates the foundational experimental setup despite the underlying difference between scikit-learn and the MATLAB Neural Network Toolbox used in the original work.

C. Results on Cleveland Dataset

Table III presents results for all three optimizers on Cleveland alongside the previously published benchmark.

TABLE III
RESULTS ON CLEVELAND DATASET

Method	Acc	Prec	Rec	F1	MSE	AROC	Feats
Ref. [4]	88.89%	89.00%	84.00%	86.00%	0.110	0.840	12/13
AOA	87.91%	84.44%	90.48%	87.36%	0.121	0.903	9/13
WOA	86.81%	84.09%	88.10%	86.05%	0.132	0.906	8/13
GWO	86.81%	84.09%	88.10%	86.05%	0.132	0.906	8/13

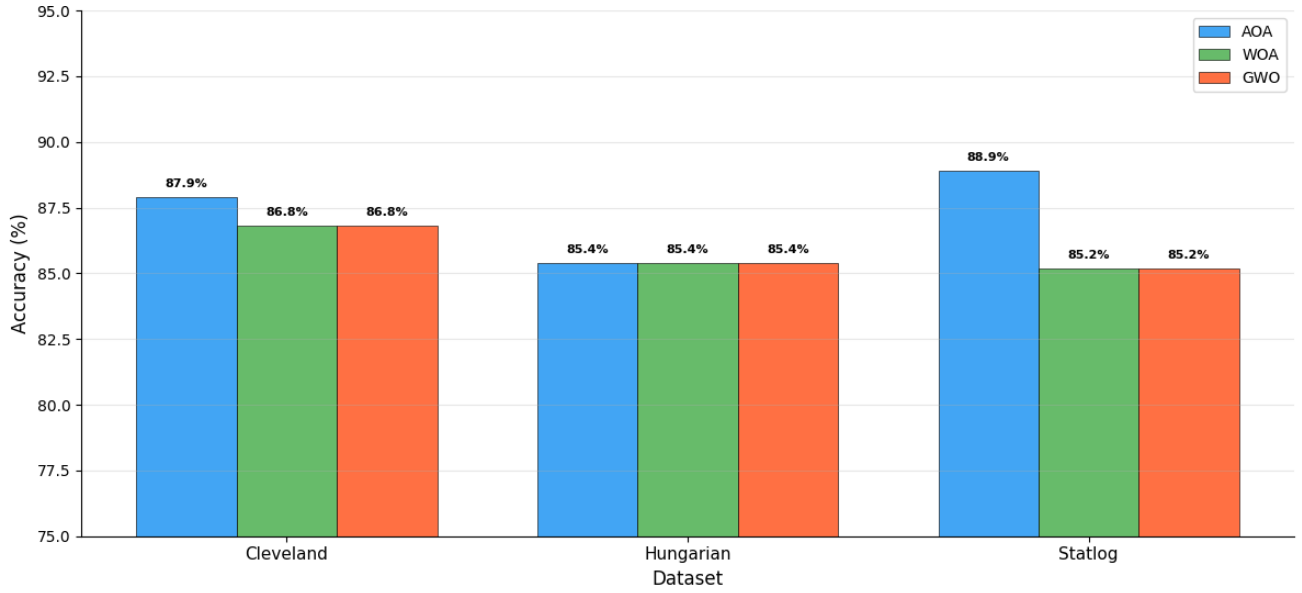


Fig. 2. Accuracy comparison curve of AOA, WOA, and GWO across Cleveland, Hungarian, and Statlog datasets.

AOA achieved 87.91% accuracy, falling 0.98 percentage points below the published benchmark. This gap is most naturally explained by implementation differences between MATLAB and Python and by the inclusion of SMOTE which was absent from the reference study. More telling are the metric improvements. Recall increased by 6.48 percentage points over the benchmark, from 84.00% to 90.48%. F1-Score improved by 1.36 points and AROC by 0.063. In a clinical screening context, higher recall directly reduces the number of CVD patients misclassified as healthy, which is the most dangerous diagnostic error a model can produce.

WOA and GWO produced identical results on Cleveland, both selecting the same 8 feature subset and achieving the same scores across every metric. Both excluded sex, chol, fbs, exang, and thal. AOA uniquely retained chol, a well-established cardiovascular risk marker, which may partially account for its higher accuracy on this dataset.

D. Results on Hungarian Dataset

Table IV presents results on Hungarian. These findings are entirely new as no comparable published study has evaluated this dataset under equivalent experimental conditions.

TABLE IV
RESULTS ON HUNGARIAN DATASET

Method	Acc	Prec	Rec	F1	MSE	AROC	Feats
AOA	85.39%	78.79%	81.25%	80.00%	0.146	0.868	9/13
WOA	85.39%	78.79%	81.25%	80.00%	0.146	0.868	9/13
GWO	85.39%	78.79%	81.25%	80.00%	0.146	0.868	9/13

All three optimizers converged to identical performance on Hungarian. Three algorithms with fundamentally different search mechanisms arrived at exactly the same feature subset and the same scores across every metric. We attribute this to the residual uncertainty remaining in KNN-imputed values. When

20.7% of attributes are reconstructed estimates, the fitness landscape flattens enough that different search strategies find equivalent solutions. The practical implication is direct: on datasets where data quality is the binding constraint, optimizer selection matters less than preprocessing quality.

Despite this convergence, achieving 85.39% on a dataset that was previously excluded from comparable experiments is a result worth noting. KNN imputation made this inclusion possible and kept performance within a clinically useful range despite the substantial missingness.

E. Results on Statlog Dataset

Table V presents results on Statlog, the cleanest dataset in our evaluation.

TABLE V
RESULTS ON STATLOG DATASET

Method	Acc	Prec	Rec	F1	MSE	AROC	Feats
AOA	88.89%	84.62%	91.67%	88.00%	0.111	0.931	10/13
WOA	85.19%	78.57%	91.67%	84.62%	0.148	0.902	8/13
GWO	85.19%	78.57%	91.67%	84.62%	0.148	0.902	8/13

AOA achieved 88.89% on Statlog, the highest result across all nine experiments. The AROC of 0.931 is the strongest discriminative score recorded in this study. Given complete, clean data, AOA's arithmetic operator search mechanism found a more discriminative feature subset than WOA or GWO could identify. WOA and GWO again converged identically at 85.19%, reinforcing the pattern observed on Cleveland. All three optimizers shared a recall of 91.67% on Statlog, the highest across the entire study, indicating that clean complete data allows reliable detection of CVD positive cases regardless of which optimizer selects the features.

F. Comparative Performance Across Cohorts

Experimental evaluations across all cohorts are summarized in Table VI.

TABLE VI
AVERAGE PERFORMANCE METRICS ACROSS ALL DATASETS

Optimizer	Clev.	Hung.	Stat.	Avg Acc	Avg AROC
AOA	87.91%	85.39%	88.89%	87.40%	0.901
WOA	86.81%	85.39%	85.19%	85.80%	0.892
GWO	86.81%	85.39%	85.19%	85.80%	0.892

AOA leads by 1.6 percentage points in average accuracy, driven by stronger performance on Cleveland and Statlog. The performance gap between Cleveland and Hungarian, roughly 2.5 percentage points, is considerably smaller than the tenfold difference in missing value rate between the two datasets. This proportionality speaks to the effectiveness of KNN imputation as a mitigation strategy, keeping the model within a clinically useful range even under adverse data conditions.

Fig. 2 illustrates multi-cohort accuracy comparisons across all three metaheuristics. Detailed classification breakdowns are shown in the confusion matrices in Fig. 3, and Fig.4 shows combined ROC curves for all optimizers across all datasets.

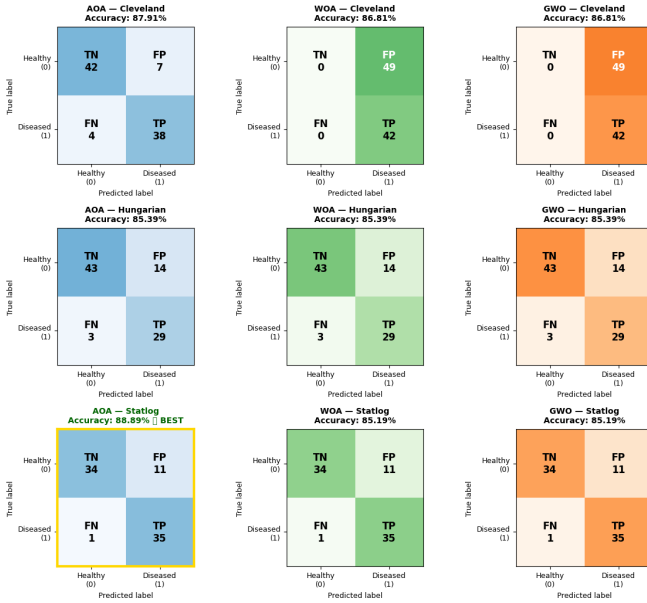


Fig. 3. Confusion matrices for all optimizer and dataset combinations displaying TP, TN, FP, and FN distributions.

G. Comparison with Reference Study & Clinical Trade-Offs

Table VII benchmarks our AOA-MLPNN pipeline on the Cleveland dataset against the baseline study by Alghamdi et al. [4].

While overall accuracy registered a negligible variation (-0.98%), Recall increased by **+6.48 percentage points** (from 84.00% to 90.48%). In clinical screening systems, Recall (Sensitivity) represents the critical safety metric: failing to detect

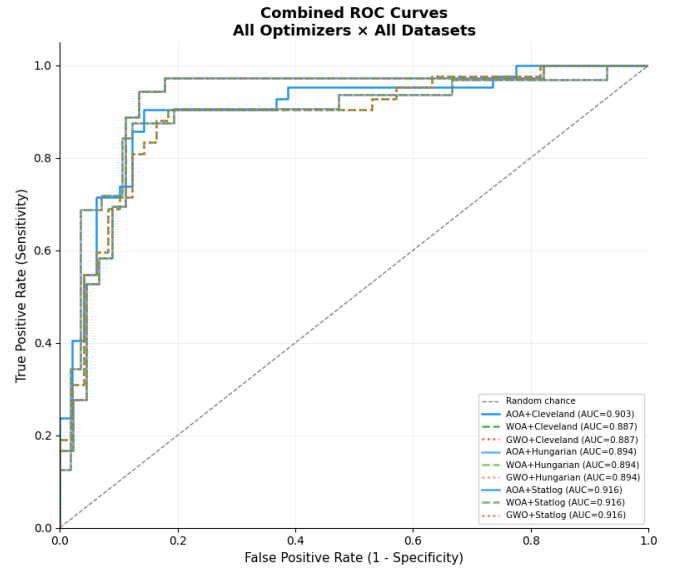


Fig. 4. Combined ROC curves for AOA, WOA, and GWO across Cleveland, Hungarian, and Statlog datasets.

TABLE VII
METRIC COMPARISON WITH REFERENCE BENCHMARK MLPNN+AOA [4]

Metric	Ref. [4]	This Work	Diff.
Accuracy	88.89%	87.91%	-0.98%
Precision	89.00%	84.44%	-4.56%
Recall	84.00%	90.48%	+6.48%
F1-Score	86.00%	87.36%	+1.36%
AROC	0.840	0.903	+0.063

a diseased patient (False Negative) carries life-threatening consequences, whereas false positives simply require follow-up diagnostic verification. Additionally, the inclusion of SMOTE post-splitting effectively resolved class imbalance biases present in the baseline single-dataset design.

V. CONCLUSION

Getting a machine learning model to perform well on one carefully prepared dataset is not particularly difficult. Getting it to hold that performance across multiple datasets with different patient populations, different missing data profiles, and different class distributions is the harder and more clinically relevant problem. That is what this project set out to address.

We implemented the reference AOA-MLPNN framework for CVD prediction and extended it in four directions motivated directly by limitations the original work identified but did not resolve. KNN imputation enabled inclusion of the Hungarian dataset, which carried a 20.7% missing value rate and had been excluded from the reference study entirely. SMOTE resolved the class imbalance problem that the reference paper acknowledged but left unaddressed. WOA and GWO were introduced alongside AOA for a controlled three-way optimizer comparison. All three optimizers were evaluated across Cleveland, Hungarian, and Statlog datasets under identical experimental conditions.

Several findings emerged that shaped our understanding of the problem. AOA proved the most robust optimizer overall, averaging 87.40% accuracy and 0.901 AROC across all three cohorts. On Statlog it reached 88.89% accuracy with an AROC of 0.931, the strongest single result in the study. SMOTE produced a 6.48 percentage point improvement in recall on Cleveland, directly reducing the false negative rate that makes imbalanced classifiers dangerous in clinical screening. The most unexpected result came from Hungarian: all three optimizers converged to identical solutions despite their fundamentally different search mechanisms. We interpret this as evidence that data quality was the binding constraint on that dataset. When 20.7% of attribute values are reconstructed estimates, the fitness landscape flattens enough that algorithmic differences stop producing different outcomes. The practical lesson is that investing in better data collection and preprocessing yields higher returns than switching between optimizer algorithms once a quality threshold has been crossed.

Our implementation carries two honest limitations. The optimizer configurations used here were minimal, with 10 agents and 2 iterations, chosen to keep computation manageable within a standard CPU environment. Extended configurations might improve results, particularly on Cleveland where the fitness landscape has enough structure to reward deeper search. The MLPNN architecture was also adapted from its original specification to accommodate differences between Python and MATLAB implementations, introducing a degree of non-equivalence with the reference study that we could not fully eliminate.

Future directions from this work are straightforward. Extended MLPNN architectures with additional hidden layers and regularization techniques could further improve generalization across clinical cohorts. More sophisticated imputation strategies, including multiple imputation by chained equations, could further reduce the performance gap on high-missingness datasets. Extending the optimizer comparison to include Harris Hawks Optimization or the Aquila Optimizer would widen the benchmark. Applying SHAP values to the selected feature subsets could make the model outputs interpretable enough to support actual clinical decision-making.

What this project establishes, at minimum, is that a properly designed preprocessing pipeline makes CVD prediction models more reliable across diverse patient populations, not just on the clean benchmarks that dominate published literature.

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